

```
1) #include <stdio.h>

int main() {
    int n, x, i, found = 0;

    scanf ("%d", &n);

    int roll [n];
    for (i = 0; i < n; i++) {
        scanf ("%d", &roll[i]);
    }

    scanf ("%d", &x);

    for (i = 0; i < n; i++) {
        if (roll[i] == x) {
            printf ("Found at index %d", i);
            found = 1;
            break;
        }
    }

    if (found == 0) {
        printf ("Not found");
    }

    return 0;
}
```

2)

#include <stdio.h>

int main() {

int n, x;

scanf("%d", &n);

int a[n];

for (int i = 0; i < n; i++)

scanf("%d", &a[i]);

scanf("%d", &x);

int l = 0, h = n - 1;

while (l <= h) {

int m = (l + h) / 2;

if (a[m] == x) {

printf("Product Available");

return 0;

}

else if (a[m] < x)

l = m + 1;

else

h = m - 1;

}

printf("Product Unavailable");

return 0;

}

3) #include <stdio.h>

int main() {

int n, temp;

scanf("%d", &n);

int a[n];

for (int i = 0; i < n; i++)

scanf("%d", &a[i]);

for (int i = 0; i < n - 1; i++) {

for (int j = 0; j < n - i - 1; j++) {

if (a[j] > a[j+1]) {

temp = a[j];

a[j] = a[j+1];

a[j+1] = temp;

}

}

for (int i = 0; i < n; i++)

printf("%d", a[i]);

return 0;


```
4) #include <stdio.h>
int main() {
    int n, key, j;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (int i = 1; i < n; i++) {
        key = a[i];
        j = i - 1;

        while (j >= 0 & a[j] > key) {
            a[j+1] = a[j];
            j--;
        }

        a[j+1] = key;
    }

    for (int i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

5) #include <stdio.h>
int main() {
 int n, min, temp;
 scanf("%d", &n);

int a[n];

for (int i = 0; i < n; i++)
 scanf("%d", &a[i]);

for (int i = 0; i < n - 1; i++) {
 min = i;

for (int j = i + 1; j < n; j++) {
 if (a[j] < a[min])
 min = j;
}

temp = a[i];
a[i] = a[min];
a[min] = temp;
}

for (int i = 0; i < n; i++)
 printf("%d", a[i]);

return 0;

}

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i, j, minIndex, temp;
6      int arr[1000];
7
8      scanf("%d", &n);
9
10     for(i = 0; i < n; i++)
11     {
12         scanf("%d", &arr[i]);
13     }
14
15     for(i = 0; i < n - 1; i++)
16     {
17         minIndex = i;
18
19         for(j = i + 1; j < n; j++)
20         {
21             if(arr[j] < arr[minIndex])
22             {
23                 minIndex = j;
24             }
25         }
26
27         temp = arr[i];
28         arr[i] = arr[minIndex];
29         arr[minIndex] = temp;
30     }
31
32     for(i = 0; i < n; i++)
33     {
```

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n, i, j, key;
6      int arr[1000];
7
8      scanf("%d", &n);
9
10     for(i = 0; i < n; i++)
11     {
12         scanf("%d", &arr[i]);
13     }
14
15     for(i = 1; i < n; i++)
16     {
17         key = arr[i];
18         j = i - 1;
19
20         while(j >= 0 && arr[j] > key)
21         {
22             arr[j + 1] = arr[j];
23             j--;
24         }
25
26         arr[j + 1] = key;
27     }
28
29     for(i = 0; i < n; i++)
30     {
31         printf("%d", arr[i]);
32
33         if(i < n - 1)
```

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  /* Function to perform binary search */
5  int binarySearch(int arr[], int n, int target) {
6      int left = 0;
7      int right = n - 1;
8      int mid;
9
10     while (left <= right) {
11         mid = left + (right - left) / 2;
12
13         if (arr[mid] == target) {
14             return 1; /* Product found */
15         }
16
17         if (arr[mid] < target) {
18             left = mid + 1;
19         } else {
20             right = mid - 1;
21         }
22     }
23
24     return 0; /* Product not found */
25 }
26
27 int main() {
28     int n, target, i;
29     int *arr;
30
31     /* 1. Read the size of the array */
32     scanf("%d", &n);
33
```

INPUT | OUTPUT | ERROR

1 Product Not Available

2



```
1 #include <stdio.h>
2
3 int linearSearch(int arr[], int size, int key)
4 {
5     int i;
6
7     for(i = 0; i < size; i++)
8     {
9         if(arr[i] == key)
10         {
11             return i;
12         }
13     }
14
15     return -1;
16 }
17
18 int main()
19 {
20     int n;
21     int rollNumbers[1000];
22     int searchRoll;
23     int indexPosition;
24     int i;
25
26     printf("Enter number of students: ");
27     scanf("%d", &n);
28
29     printf("Enter roll numbers:\n");
30     for(i = 0; i < n; i++)
31     {
32         scanf("%d", &rollNumbers[i]);
33     }
34
35     printf("Enter roll number to search: ");
36     scanf("%d", &searchRoll);
37
38     indexPosition = linearSearch(rollNumbers, n, searchRoll);
39
40     if(indexPosition != -1)
41     {
42         printf("Found at index %d", indexPosition);
43     }
44     else
45     {
46         printf("Not Found");
47     }
48
49     return 0;
50 }
```

Enter number of students: 5

Enter roll numbers:

1

2

3

4

5

Enter roll number to search: 4

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=== Code Execution Successful ===